

RIDOPANDI SINAGA

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience developing AI solutions using generative AI, machine learning, and deep learning. Skilled in Python, scikit-learn, TensorFlow, LangChain, and LangGraph, with experience in data processing, model development and evaluation, and end-to-end agentic RAG applications. Combines AI development with API integration and system design to build reliable solutions for real-world business needs.

WORK EXPERIENCE

AI Developer - Freelance

Optik 150 K

Aug 2025 - Oct 2025

- **AI Customer Service & Agentic RAG:** Built an end-to-end AI customer service agent using LangGraph and RAG to support product discovery, recommendations, price calculation, order creation, and payment assistance.
- **Human-in-the-Loop & Operations Dashboard:** Designed an operations dashboard for staff to monitor conversations, approve sensitive actions, take over customer chats, and return control to the AI while preserving conversational context.
- **Payment Reliability & AI Observability:** Integrated Midtrans QRIS with webhooks, polling, persistent state, and idempotency; implemented Langfuse to monitor LLM traces, token usage, costs, latency, tool execution, and RAG quality.

AI Developer - Contract

PT Riset Perkebunan Nusantara

Feb 2025 - May 2025

- **OCR & Knowledge Engineering:** Built an automated OCR and document processing pipeline for **100+** unstructured commercial books, converting heterogeneous content into a structured, searchable, and traceable AI knowledge base.
- **RAG Chain Development:** Built RAG chains from scratch with LangChain and Qdrant, covering document parsing, chunking, embedding, vector indexing, retrieval, context construction, and LLM integration for document-grounded answers.
- **Retrieval Optimization & Reliability:** Refined chunking, metadata, retrieval settings, prompts, and chain orchestration; evaluated retrieved context and LLM responses to improve relevance, multi-document search quality, and system reliability.

AI Engineer Intern

Indonesian Oil Palm Research Institute (IOPRI)

Sep 2024 - Dec 2024

- Improved chatbot Context Recall by **12%** by implementing a Neo4j knowledge graph and refining vector and GraphRAG retrieval across internal product information.
- Developed and evaluated a lightweight language model for structured Cypher query generation to support knowledge graph-based information retrieval.
- Evaluated retrieval and response quality using RAGAS, identifying failure cases and supporting iterative improvements to system accuracy and reliability.

EDUCATION & DEVELOPMENT

Universitas Sumatera Utara

2021 - 2026

Bachelor of Computer Science, majoring in Information Technology | GPA: 3.92/4.00

Thesis: Implemented and evaluated agentic behavior in LLM-powered RAG pipelines by comparing traditional RAG with single-agent and multi-agent approaches using DeepSeek-V3. Evaluated retrieval and response quality using RAGAS and Wilcoxon signed-rank tests, increasing Context Recall from **73.60%** to **85.60%** with statistically significant improvement ($p < 0.001$) and achieving **92.16%** Answer Relevancy, with the strongest gains observed in multi-document retrieval scenarios.

Bangkit Academy led by Google, GoTo, and Traveloka

Feb 2024 - Jul 2024

Selected as one of **4,650** participants from **55,000** applicants nationwide (approximately **8.5%** acceptance rate) for Bangkit Academy 2024 Batch 1. Completed the Machine Learning Learning Path through an industry-aligned intensive program covering Python-based data analysis, IT automation, machine learning, TensorFlow model development, and deployment.

ORGANIZATION EXPERIENCE

Google Developer Student Clubs (GDSC) USU - Staff of Technical Division

Sep 2023 - Jul 2024

- Organized subject matter to design an interactive curriculum for the GDSC USU period, resulting in increased student satisfaction and a **22%** decrease in low-interest rates.
- Established connections with experienced speakers and secured their participation in workshops and events, resulting in a **35%+** increase in workshop registrations and **15%+** growth in online RSVPs.

AWARD

3rd Place Winner - Fordigi Hackathon Challenge 2023, City Chapter Medan

Placed **third** in the Medan chapter of a nationwide hackathon across **ten** Indonesian cities by co-developing Parentify, an AI-powered parenting education prototype addressing local health and hygiene challenges.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Data Processing & Cleaning, Exploratory Data Analysis

Software, Systems & Database: FastAPI, REST APIs, MySQL, Docker, Git, Model Deployment, Qdrant, Neo4j

Machine Learning & AI: Machine Learning, scikit-learn, TensorFlow, NLP, Information Retrieval, RAG, GraphRAG, Fine-tuning, LLM/Model Evaluation, Statistical Testing, RAGAS, LangChain, LangGraph, ReAct Agent, Langfuse

Certifications: Machine Learning Specialization · Process Data from Dirty to Clean · TensorFlow Developer Professional Certificate